

DOFBOT Up Down Left Right

1. Experiment Approach

This experiment controls the DOFBOT to swing up, down, left, and right, then returns to the upright state. By simultaneously controlling servos 3 and 4 at different angles, we achieve the up/down swinging function. Then we control servo 1 for left/right swinging, and finally return to the upright state.

2. Code Content

Code path: /home/yahboom/Dofbot/3.ctrl_Arm/6.left_right.ipynb

```
#!/usr/bin/env python3
#coding=utf-8
import time
from Arm_Lib import Arm_Device
# Create DOFBOT object
Arm = Arm_Device()
time.sleep(.1)
```

```
def main():
    # Reset servos to center position
    Arm.Arm_serial_servo_write6(90, 90, 90, 90, 90, 90, 500)
    time.sleep(1)
    while True:
        # Control servos 3 and 4 to move up and down
        Arm.Arm_serial_servo_write(3, 0, 1000)
        time.sleep(.001)
        Arm.Arm_serial_servo_write(4, 180, 1000)
        time.sleep(1)

        # Control servo 1 to move left and right
        Arm.Arm_serial_servo_write(1, 180, 500)
        time.sleep(.5)
        Arm.Arm_serial_servo_write(1, 0, 1000)
        time.sleep(1)

        # Reset servos to initial position
        Arm.Arm_serial_servo_write6(90, 90, 90, 90, 90, 90, 1000)
        time.sleep(1.5)
    try :
        main()
    except KeyboardInterrupt:
        print(" Program closed! ")
        pass
```

```
del Arm # Release Arm object
```

